Furey Fermions in GALG and FFGFT:

N-Operator, Tripotents and Neutrino-Vacuum Identity

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Abstract

Doug Matzke's IPI mail of 3 September 2026 presents concrete GALG results on Furey fermions in Cl(6): complete Up/Down fermion sets Su/Sd, idempotents N_1, N_2, N_3, tripotent N, jE7 projector, and a grade-1+grade-5 construction. The strongest single result: Su$[0]$ = Vss$[0]$ exactly — the neutrino is the lowest vacuum state **[B]**. Theorem F shows: the tripotent property for even vacua and the Frobenius fixed-point property (Doc. 339) are the same decomposition from two independent directions **[B]**. The electron exception at the jE7 projector is the known Furey problem **[S]**.

1 Theorem A — N_k are idempotents in characteristic 3

$N_k = -1 + (-i\,B_k)$, $B_k^2 = -1$ (Witt pair):

$$N_k^2 = 1 + 2i\,B_k + (-1)(-1) = 2 + 2i\,B_k \equiv -1 + (-i\,B_k) \pmod 3 = N_k. \quad \textbf{[B]}$$

The sum loses its scalar part since $-3 \equiv 0$: $N_1 + N_2 + N_3 \equiv (-i) \sum B_k = N$.

2 Theorem B — N is a tripotent

Cross-terms $N_k N_j$ $(k \neq j)$ are pure grade-4 elements; hence $N^3 = N$ **[B]**. Doug's output: `N**3 = N`, `N**4 = N**2`.

3 Theorem C — Su$[0]$ = Vss$[0]$: neutrino is the vacuum

`Su[0] == Vss[0]: True` and `(1+N)*(1+jE7) == Vss[0]: True` **[B]**.

FFGFT correspondence [B]. In FFGFT, ν_1 lies in orbit $O_1 = \{1, 3, 9\}$, the orbit of the generator of GF$(27)^*$ — the algebraically simplest object. Both frameworks independently place the neutrino at the algebraically simplest position: convergence of two independent frameworks.

Why are the neutrino and the vacuum the same?

Three independent routes lead to the same result:

1. Furey/GALG (this theorem). Vacuum state Vss[0] is defined as the object annihilated by all creation operators: no Fock-space content, no quantum numbers. The neutrino Su[0] has the same algebraic structure: no colour, no electric charge, only minimal weak coupling. Hence they are identical.

2. Galois/FFGFT (Doc. 339/343). GF$(27)^*$ has a generator g of order 26. Orbit $O_1 = \{g^1, g^3, g^9\}$ is the orbit of the generator itself — the algebraically simplest object in the structure. ν_1 lies in O_1. The vacuum Vss[0] is the state with zero torus winding ($n = 0$). Smallest mass ↔ simplest orbit ↔ $n = 0$ state — three names for the same minimality property.

3. Photon analogy (Doc. 007). $m_\nu = \xi^2/2 \times m_e = 4.54$ meV. The photon has mass 0 and is the limiting case of the vacuum. The neutrino is the massive particle closest to the vacuum: double ξ suppression = maximum approach to $m = 0$ among all massive fermions **[K]**.

Structural statement. In a theory that derives all particles from an algebraic structure, the lightest massive fermion must lie closest to the vacuum state. Neutrino and vacuum are the algebraic minimum: the same object seen from different directions **[B]**.

Remark on the vacuum identification. Doug Matzke works in Cl(6) over GF(9) without a mass scale — in GALG mass does not appear. When he identifies Su[0] = Vss[0] he means: same algebraic structure, no mass value behind it. The neutrino is the "most neutral" fermion of the algebra, the vacuum is the "emptiest" state — neither carries any of the quantum numbers that distinguish other objects.

In FFGFT the step is more precise: Vss[0] corresponds to the $n = 0$ state on the torus; the Galois orbit O_1 assigns it the mass $m_{\nu_1} = 4.54$ meV (Doc. 340). The absolute mass in meV arises through the SI energy scale: ξ is derived from Galois group orders **[K]**, the torus quantum numbers (r_e, p_e) give m_e parameter-free **[K]**, and $v = 246$ GeV is the SI unit choice (no free parameter, but a scale fixation). Doug sets no units, so this step does not exist in GALG. The identification Su[0] = Vss[0] is algebraically exact **[B]**; in FFGFT it means "ground state with smallest mass", not "massless object". The absolute value 4.54 meV is parameter-free: ξ **[K]** (Galois group orders, Doc. 338), m_e **[K]** (torus quantum numbers, Doc. 006), $v = 246$ GeV = SI scale fixation (not a free constant).

4 Theorems D–F

D — sigma [B]. $e_1 \to e_2 \to e_4 \to e_1$, $e_3 \to e_6 \to e_5 \to e_3$; `{1:2,2:4,4:1,3:6,6:5,5:3}`, two 3-cycles, $\sigma^3 =$ id. Doc. 341 contains this formula; Doug's `all_g1_g5good` is the σ-invariant sum.

E — jE7 ≡ N_k as projectors [B]; electron exception [S]. $F \cdot jE7 = +F$ for all Su (8/8); $F \cdot jE7 = -F$ for 7/8 of Sd (Sd[7]=electron fails). Same for N_k. Known Furey problem. FFGFT: electron in Dirac layer $\{f_3, f_4\}$, neutrino in Majorana layer $\{f_1, f_2\}$ (Doc. 343, Theorem D‴). Common root open **[S]**.

F — Tripotent for even vacua [B]. $(V_k + P)^2 = -V_k$, $(V_k + P)^3 = V_k$ for $k \in \{0, 2, 4\}$; no collapse for $k \in \{1, 3, 5\}$ (length 32). Identical to the Frobenius fixed-point/orbit split of Doc. 339 — two independent derivations of the same decomposition.

5 Theorem G — Anti-neutrinos: Dirac or Majorana?

The tension. In GALG/Furey (Theorem C) the neutrino is Su$[0]$ = Vss$[0]$ and the anti-neutrino is Sd$[0]$ = cc(Su$[0]$) = Vss$[1]$ ≠ Vss$[0]$. Hence neutrinos in Furey's construction are **Dirac particles** ($\nu \neq \bar{\nu}$).

In FFGFT (Doc. 343, Theorem D‴), ν_1 and ν_2 lie in orbits O_1 and O_3, which are self-inverse under $k \mapsto k^{-1}$ — the *algebraic* Majorana criterion.

Theorem 5.1 (Clarification of Majorana concept **[B]**/**[S]**). *Two distinct Majorana concepts:*

1. ***Algebraic [B]:*** *Orbit O_k is self-inverse under $k \mapsto k^{-1}$ mod 13 — a statement about GF$(27)^*$ structure, not about spin.*
2. ***Physical [S]:*** *$\nu = \bar{\nu}$ in the Lorentz sense (Majorana condition). This requires Sd[0]=Su[0], which does not hold in GALG/Furey.*

*The Galois algebra fixes the algebraic property **[B]**; the physical Majorana condition is an additional assumption **[S]**.*

Position of anti-neutrinos [B]/[S]. GALG: $\bar{\nu}$ = Sd[0] = Vss[1] in the odd vacuum (sigma-orbit, other sector, Doc. 339). FFGFT: the anti-neutrino would lie in the negation $f_1 \leftrightarrow g_1$ (Doc. 343, Theorem D″) — order-13 class g_1, other sector. Both frameworks place the anti-neutrino in the structurally "other" sector.

Consequence for m_{ee}. The prediction $m_{ee} \approx 7.1$ meV from Doc. 340 assumes physical Majorana neutrinos $\nu_{1,2}$. If ν_1, ν_2 are instead Dirac particles (as in Furey), m_{ee} vanishes entirely. The m_{ee} prediction is conditioned on the Majorana assumption **[S]**.

Open question [S]. Common algebraic root of the electron inconsistency in GALG/Furey and the Majorana/Dirac layer in FFGFT.

Empirical decision. nEXO (sensitivity $m_{\beta\beta} \approx 5$–20 meV depending on nuclear matrix element) decides: signal at ~ 7 meV ⇒ Majorana (Doc. 340 confirmed, Theorem H refuted); no signal ⇒ consistent with Dirac (Theorem H confirmed or destructive interference).

6 Theorem H — The Galois algebra forces Dirac neutrinos

Theorem 6.1 (Dirac constraint **[B]** under assumption **[S]**). *If the sector pairing $k \mapsto -k$ in GF$(27)^*$ corresponds to the physical charge conjugation C, then **all** neutrinos are Dirac particles.*

Proof. A Majorana particle is a fixed point of charge conjugation: $\nu = \bar{\nu}$, i.e. class f_i must map to itself under C.

Under the sector pairing $k \mapsto -k$ (mod 26):

$$\begin{aligned} f_1 = \{1,3,9\} &\;\mapsto\; \{25,23,17\} = f_2 \neq f_1 \\ f_2 = \{17,23,25\} &\;\mapsto\; f_1 \neq f_2 \\ f_3 = \{5,15,19\} &\;\mapsto\; f_4 \neq f_3 \\ f_4 = \{7,11,21\} &\;\mapsto\; f_3 \neq f_4 \end{aligned}$$

No primitive class is a fixed point of the sector pairing **[B]**. A Majorana fixed point would require $f_i \mapsto f_i$ — this does not exist in GF$(27)^*$. The anti-particle of the neutrino f_1 is f_2: $\nu_1 \neq \bar{\nu}_1$. □ □

Assumption [S]. The argument holds if $k \mapsto -k$ in GF$(27)^*$ corresponds to physical charge conjugation C. In Doc. 336 the Frobenius involution $x \mapsto x^3$ is identified with the sector separation; the identification with C is physically motivated there but not algebraically proved.

Consistency [B]. In GALG/Furey (Doc. 344, Theorem C): Sd$[0]$ = Vss$[1]$ ≠ Vss$[0]$ = Su$[0]$, i.e. $\bar{\nu} \neq \nu$ — fully consistent with this theorem.

Plain language: What is m_{ee} and why does it matter?

The process. A radioactive nucleus converts two neutrons into two protons simultaneously, emitting two electrons. In ordinary double beta decay ($2\nu\beta\beta$) two anti-neutrinos also come out — the energy is shared among four particles. In *neutrinoless* decay ($0\nu\beta\beta$) no anti-neutrinos come out.

Why this requires Majorana. The anti-neutrino produced at the first nuclear vertex must be absorbed as a neutrino at the second vertex. This is only possible if neutrino and anti-neutrino are the same particle ($\nu = \bar{\nu}$, Majorana condition). For Dirac neutrinos ($\nu \neq \bar{\nu}$) the process is exactly forbidden.

What m_{ee} measures. The half-life of $0\nu\beta\beta$ decay is proportional to $1/m_{ee}^2$, where

$$m_{ee} = \left|m_1 U_{e1}^2 + m_2 U_{e2}^2 + m_3 U_{e3}^2\right|$$

combines all three neutrino masses weighted by PMNS mixing elements and Majorana phases.

Why interference matters. The three terms can add (constructively) or partially cancel (destructively) depending on the Majorana phases. With FFGFT masses from Doc. 340 and phases ≈ 0: constructive gives $m_{ee} \approx 7.1$ meV; destructive gives $m_{ee} \approx 0.12$ meV; Dirac gives $m_{ee} = 0$ exactly.

What nEXO can measure. nEXO targets $m_{\beta\beta} \approx 5$–20 meV (NME-dependent, nEXO 2022); the FFGFT value 7.1 meV lies in the decision range. A signal at ~ 7 meV confirms Majorana and refutes Theorem H; no signal down to 5 meV excludes the constructive Majorana prediction and is consistent with Dirac or destructive interference.

Consequence: $m_{ee} = 0$ and no neutrinoless double beta decay

For Dirac neutrinos, neutrinoless double beta decay ($0\nu\beta\beta$) is **exactly forbidden**:

$$m_{ee} = \left|\sum_i m_i U_{ei}^2\right| \equiv 0.$$

Table 1: Predictions: Majorana (Doc. 340) vs. Dirac (Theorem H)

	Majorana ν_1, ν_2 **[S]**	Dirac all **[B]**
m_{ee}	≈ 7.1 meV	$= 0$
$0\nu\beta\beta$	allowed, ~ 10 meV	exactly forbidden
nEXO signal	~ 7 meV at phases ≈ 0	no signal
KamLAND-Zen	consistent (< 28–122 meV, 2025)	consistent

Testability. nEXO ($m_{\beta\beta} \approx 5$–20 meV) and the complete KamLAND-Zen-800 dataset (28–122 meV, PRL 135 (2025) 262501) bracket the range. A null result at nEXO excludes Majorana with Doc. 340 masses (phases ≈ 0) and is consistent with Theorem H. A signal at ~ 7 meV refutes Theorem H and confirms the Majorana interpretation **[B]**/**[S]** of Doc. 340.

What we know better since Galois

Why exactly the primes {5,7,11,13} — [K]

Before: phenomenological observation. Now: algebraic theorem. They are the only primes entering $|\mathrm{GF}(3^k)^*|$ at $k = 3, 4, 5, 6$. $p = 13$ is the worldwide unique prime with $\mathrm{ord}_p(3) = 3$. This follows necessarily from the group structure — no choice (Doc. 342, R106).

Where $\xi = 4/30000$ comes from — [K]

Before: defined, not derived. Now: ξ follows algebraically from the identity $(r_\mu/r_e)^2/\xi = 43200 = |\mathrm{GF}(9)^*|^2 \cdot 5^2 \cdot |\mathrm{GF}(27)|$ and the torus winding numbers $r_\mu/r_e = 12/5$. ξ is not a free constant (Doc. 338, R101).

Why the neutrino mass is 4.54 meV — [K]

Before: from $\xi^2/2 \times m_e$ (Doc. 007), without explanation of origin. Now closed: ξ **[K]** (Galois group orders, Doc. 338), m_e **[K]** (torus quantum numbers $n = 1, \ell = 0, j = \frac{1}{2}$, Doc. 006), $v = 246$ GeV = SI scale fixation (not a free parameter, but a unit choice). The 4.54 meV follow without any free constant — parameter-free.

Why exactly two fermion layers — [B]

Before: observation. Now: the sector pairing $k \mapsto -k$ on $\mathrm{GF}(27)^*$ has no fixed points among the primitive classes — it necessarily pairs $f_1 \leftrightarrow f_2$ and $f_3 \leftrightarrow f_4$. Four classes become exactly two pairs. This is a group-theoretic theorem (Doc. 343, Theorem D).

Why the neutrino is the lightest massive fermion — [B]

Before: no algebraic explanation. Now: $O_1 = \{1, 3, 9\}$ is the orbit of the generator of $\mathrm{GF}(27)^*$ — the algebraically simplest element. The neutrino lies there. Simultaneously: $\mathrm{Su}[0] = \mathrm{Vss}[0]$ in GALG/Furey (Theorem C) — the neutrino is the vacuum state of the Clifford Fock space. Three independent routes converge **[B]**.

A sharp open question instead of a vague one — [B]/[S]

Before: unclear whether neutrinos are Majorana or Dirac. Now precise: the sector pairing has no fixed point ⇒ no Majorana possible under assumption $k \mapsto -k = C$ (Theorem H) **[B]/[S]**. If yes: $m_{ee} = 0$ exactly, $0\nu\beta\beta$ forbidden. If no: $m_{ee} \approx 7.1$ meV. nEXO decides at 5–20 meV sensitivity (NME-dependent).

Why there are no further particles — [B]

This is one of the strongest statements: the algebra prescribes not only what exists, but also what cannot exist.

Four primitive classes, no more. $\mathrm{GF}(27)^*$ has exactly $\varphi(26) = 12$ primitive elements forming exactly 4 Frobenius three-cycles. There is no fifth primitive class.

Two layers, no more. The sector pairing $k \mapsto -k$ pairs $f_1 \leftrightarrow f_2$ and $f_3 \leftrightarrow f_4$ — exactly two pairs, no remainder. Negation $f_i \leftrightarrow g_i$ yields sign partners (order 13), not independent new particle classes.

Three generations, not four. Each primitive class is a three-cycle. 3 elements per orbit = 3 generations. Forced by the Frobenius orbit length, not chosen.

16 fermions per generation, complete (GALG/Furey). $8 + 8$ = Su ∪ Sd covers exactly one generation of the Standard Model: neutrino, 3 colours × up-quark, 3 colours × down-quark, positron (and their Down-sector partners). No gap, no room for more.

No sterile neutrinos from this algebra. The Majorana layer $\{f_1, f_2\}$ is fully occupied by ν_1 and ν_2. A third Majorana slot does not exist (Doc. 343, Theorem D‴).

What we do not know better

The electroweak scale $v = 246$ GeV is not yet forced from algebraic principles — a unit choice, not a free parameter, but its origin is open. The electron exception in GALG (jE7 fails for Sd[7]=electron) has no algebraic explanation in FFGFT yet (Theorem E, **[S]**).

7 Summary

Table 2: Results Doc. 344

Theorem	Content	Status
A	$N_k^2 = N_k$, characteristic-3 proof	**[B]**
B	$N^3 = N$, tripotent	**[B]**
C	Su[0]=Vss[0]: neutrino=vacuum; FFGFT correspondence	**[B]**
D	sigma= $\{1 \to 2, 2 \to 4, 4 \to 1, 3 \to 6, 6 \to 5, 5 \to 3\}$	**[B]**
E	jE7≡ N_k; electron exception (Furey problem)	**[B]**/**[S]**
F	Tripotent even vacua = Frobenius fixed points (Doc. 339)	**[B]**
G	Anti-neutrino Sd[0]=Vss[1]≠Su[0]; algebraic Majorana ≠ physical Majorana	**[B]**/**[S]**
H	Sector pairing $k \mapsto -k$ has no fixed point: all neutrinos Dirac; $m_{ee} = 0$; $0\nu\beta\beta$ forbidden	**[B]**/**[S]**

Open question [S]: Common algebraic root of the electron inconsistency in GALG/Furey and the Majorana/Dirac layer separation in FFGFT (Doc. 343, Theorem D‴).

Verification script

`pruef_344_furey_galg.py` — 6 theorems, all assertions passed.

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